

Unit 2

Module 2 : First Order Ordinary Differential Equations

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First Order Ordinary Differential Equations

Syllabus

Exact differential equations, Integrating factors for non-Exact differential equations, linear and Bernoulli's equations. Equations not of first degree: equations solvable for p , equations solvable for y , equations solvable for x and Clairaut's type.

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2.1 Introduction

Differential equations appear frequently in mathematical models that effort to describe real-life situations. The topics of differential equations is an extremely important one in mathematics, science and engineering as well as many other branches of studies in which changes occur and in which predictions are desirable. For example, derivatives appear in Physics as velocity and accelerations, in Biology as rates of growth of populations, in Chemistry as reaction rates, in Economics as rate of change of cost of living, in Finance as rate of growth of investments. The rate of change which does not involve any higher derivatives. It is therefore of interest to study first order differential equations.

Definition : A first order differential equation is an equation of the form $F(x, y, y') = 0$. A solution of a first order differential equation is a function $f(x)$ that makes $F(x, f(x), f'(x)) = 0$ for every value of x .

2.2 Classification of Differential Equations

Differential equations may be classified in two categories as,

- i) Ordinary differential equation.
- ii) Partial differential equation.

Ordinary differential equation :

An equation is called an ordinary differential equation, if it involves a function of a single variable and ordinary derivative of that function.

i.e.

- 1) $\frac{dy}{dx} + 2xy = 0$
- 2) $\frac{d^2y}{dx^2} + x^2y = 0$
- 3) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$

Partial differential equation :

An equation is called a partial differential equation, if it involves a function of more than one variable and partial derivatives of that function.

i.e.

- 1) $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$
- 2) $\frac{\partial^2 u}{\partial t^2} = 6^2 \frac{\partial^2 u}{\partial x^2}$
- 3) $\frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial x \partial y} + 6 \frac{\partial^2 z}{\partial y^2} = 0$

2.3 Order and Degree of Differential Equations

GTU : Winter-21

The order of highest derivative which appeared in differential equation is called the order of the differential equation.

The degree of highest order derivative of a differential equation when all the derivatives are free from radicals and fractions is called a degree of differential equations.

i.e.

- 1) $\left(\frac{d^2y}{dx^2}\right)^2 + \frac{dy}{dx} + 3y = 0$
- 2) $\left(\frac{d^3y}{dx^3}\right)^2 - 2x\left(\frac{d^2y}{dx^2}\right)^3 + y = 0$

Order is 2. Degree is 2

Order is 3. Degree is 2

- 3) $\left(1 + \frac{d^3y}{dx^3}\right)^{3/2} = \sqrt{1 - \left(\frac{dy}{dx}\right)^3}$

$$\Rightarrow \left(1 + \frac{d^3y}{dx^3}\right)^3 = 1 - \left(\frac{dy}{dx}\right)^3$$

Order is 3. Degree is 3.

Ex. 2.3.1 : Find order and degree of $\left(\frac{dy}{dx} + y\right)^{1/2} = \sin x$

GTU : Winter-21, Maths-3, Marks 2

Sol. :

$$\left(\frac{dy}{dx} + y\right)^{1/2} = \sin x$$

$$\Rightarrow \frac{dy}{dx} + y = \sin^2 x$$

$\therefore$ Order is 1. Degree is 1.

2.4 Solution of Differential Equation

A relation between the dependent and independent variables involved which satisfies the given differential equation is called a solution of that equation. The types of solutions are general solution, particular solution.

General solution :

A solution of differential equation is said to be general solution, if it contains the number of arbitrary constants is equal to the order of the given differential equation.

Particular solution :

A solution which is obtained from the general solution by giving particular values to arbitrary constant is called particular solution.

Initial Value Problem (IVP) :

A problem is said to be initial value problem if the particular solution of a differential equation is obtained subject to the condition on dependent variable and its derivative, at one value of the independent variable.

Boundary Value Problem (BVP) :

A problem is said to be boundary value problem if the particular solution of a differential equation is obtained subject to the conditions at more the one values of independent variable, then the problem is called boundary value problem.

2.5 Exact Differential Equation

GTU : Winter-18, 19, 21, 22, 23, Summer-19, 20, 22, 23, 24

A differential equation of the form $M(x, y) dx + N(x, y) dy = 0$ is called an exact differential equation if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, where first order continuous partial derivatives of M and N are exist.

Solution of exact differential equation :

If the differential equation $M(x, y)dx + N(x, y)dy = 0$ is exact then solution is

$$\int M(x, y)dx + \int N(x, y)dy = C \text{ (Constant)}$$

(Term of N)
(free from x)

Ex. 2.5.1 : Solve $(x^3 + 3xy^2)dx + (3x^2y + y^3)dy = 0$.

GTU : Summer-20, Maths-2, Marks 4, Winter-23, Maths-3

Sol. : Here, $M = x^3 + 3xy^2$, $N = 3x^2y + y^3$

$$\frac{\partial M}{\partial y} = 6xy, \quad \frac{\partial N}{\partial x} = 6xy$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is exact.

$\therefore$ Solution is

$$\int M dx + \int N dy = C$$

(Term of N)
(free from x)

$$\Rightarrow \int (x^3 + 3xy^2)dx + \int y^3 dy = C$$

$$\Rightarrow \frac{x^4}{4} + \frac{3x^2y^2}{2} + \frac{y^4}{4} = C$$

Ex. 2.5.2 : Solve $(x^2 - x - y^2)dx - 2xydy = 0$.

GTU : Summer-22, Maths-3, Marks 4

Sol. : Here, $M = x^2 - x - y^2$, $N = -2xy$

$$\frac{\partial M}{\partial y} = -2y, \quad \frac{\partial N}{\partial x} = -2y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equations is exact.

$\therefore$ Solution is

$$\int M dx + \int N dy = C$$

(Term of N)
(free from x)

$$\Rightarrow \int (x^2 - x - y^2)dx + \int 0 dy = C$$

$$\Rightarrow \frac{x^3}{3} - \frac{x^2}{2} - xy^2 = C$$

Ex. 2.5.3 : $ye^x dx + (2y + e^x)dy = 0$.

GTU : Winter-19, Maths-2, Marks 4, Summer-22, 23, Maths-3, Winter-23, AEM

Sol. : Here, $M = ye^x$, $N = 2y + e^x$

$$\frac{\partial M}{\partial y} = e^x, \quad \frac{\partial N}{\partial x} = e^x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is exact.

$\therefore$ Solution is

$$\int M dx + \int N dy = C$$

(Term of N)
(free from x)

$$\Rightarrow \int ye^x dx + \int 2y dy = C$$

$$\Rightarrow ye^x + y^2 = C$$

Ex. 2.5.4 : Solve $(x^2 - 2xy - y^2)dx - (x + y)^2 dy = 0$.

GTU : Summer-24, Maths-3, Marks 4

Sol. : Here, $M = x^2 - 2xy - y^2$, $N = -(x + y)^2$

$$\frac{\partial M}{\partial y} = -2x - 2y, \quad \frac{\partial N}{\partial x} = -2(x + y)$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is exact.

$\therefore$ Solution is,

$$\int M dx + \int N dy = C$$

(Term of N)
(free from x)

$$\Rightarrow \int (x^2 - 2xy - y^2) dx + \int (-y^2) dy = C$$

$$\Rightarrow \frac{x^3}{3} - x^2y - xy^2 - \frac{y^3}{3} = C$$

Ex. 2.5.5 : Solve $(x + y - 2)dx - (x - y + 4) dy = 0$.

GTU : Summer-19, AEM, Marks 3

Sol. : Here, $M = x + y - 2$, $N = x - y + 4$

$$\frac{\partial M}{\partial y} = 1, \quad \frac{\partial N}{\partial x} = 1$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is exact.

$\therefore$ Solution is

$$\int M dx + \int N dy = C$$

(Term of N)
(free from x)

$$\Rightarrow \int (x + y - 2)dx + \int (-y + 4)dy = C$$

$$\Rightarrow \frac{x^2}{2} + xy - 2x - \frac{y^2}{2} + 4y = C$$

Ex. 2.5.6 : Solve $\sinh x \cos y dx = \cosh x \sin y dy$.

GTU : Winter-18, AEM, Marks 4

Sol. : $\sinh x \cos y dx - \cosh x \sin y dy = 0$

Here, $M = \sinh x \cos y$,

$N = -\cosh x \sin y$

$$\frac{\partial M}{\partial y} = -\sinh x \sin y, \quad \frac{\partial N}{\partial x} = -\sinh x \sin y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is exact.

$\therefore$ Solution is

$$\int M dx + \int N dy = C$$

(Term of N)
(free from x)

$$\Rightarrow \int \sinh x \cos y dx + \int 0 dy = C$$

$$\Rightarrow \cosh x \cos y = C$$

Ex. 2.5.7 : Solve $(y^2 - x^2)dx + 2xy dy = 0$.

GTU : Winter-22, AEM, Summer-24, Maths-2, Marks 3

Sol. : Here, $M = y^2 - x^2$, $N = 2xy$

$$\frac{\partial M}{\partial y} = 2y, \quad \frac{\partial N}{\partial x} = 2y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is exact.

$\therefore$ Solution is

$$\int M dx + \int N dy = C$$

(Term of N)
(free from x)

$$\Rightarrow \int (y^2 - x^2) dx + \int 0 dy = C$$

$$\Rightarrow xy^2 - \frac{x^3}{3} = C$$

Ex. 2.5.8 : Solve $(y \cos x + 2x e^y) + (\sin x + x^2 e^y - 1)y' = 0$.

GTU : Summer-22, Maths-2, Marks 4

Sol. :

$$(y \cos x + 2x e^y) + (\sin x + x^2 e^y - 1)y' = 0$$

$$(y \cos x + 2x e^y) + (\sin x + x^2 e^y - 1) \frac{dy}{dx} = 0$$

$$(y \cos x + 2x e^y) dx + (\sin x + x^2 e^y - 1) dy = 0$$

Here,

$$M = y \cos x + 2x e^y$$

$$N = \sin x + x^2 e^y - 1$$

$$\frac{\partial M}{\partial y} = \cos x + 2x e^y$$

$$\frac{\partial N}{\partial x} = \cos x + 2x e^y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is exact.

$\therefore$ Solution is

$$\int M dx + \int N dy = C$$

$$\left(\begin{array}{l} \text{Term of N} \\ \text{free from x} \end{array} \right)$$

$$\Rightarrow \int (y \cos x + 2x e^y) dx + \int (-1) dy = C$$

$$\Rightarrow y \sin x + x^2 e^y - y = C$$

Ex. 2.5.9 : Solve $e^x \cos y dx - e^x \sin y dy = 0$.

GTU : Summer-23, Maths-2, Marks 3

Sol. : Here, $M = e^x \cos y$, $N = -e^x \sin y$

$$\frac{\partial M}{\partial y} = -e^x \sin y \quad \frac{\partial N}{\partial x} = -e^x \sin y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is exact.

$\therefore$ Solution is

$$\int M dx + \int N dy = C$$

$$\left(\begin{array}{l} \text{Term of N} \\ \text{free from x} \end{array} \right)$$

$$\Rightarrow \int e^x \cos y dx + \int 0 dy = C$$

$$\Rightarrow e^x \cos y = C$$

Ex. 2.5.10 : Solve $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$.

GTU : Winter-21, Maths-2, Marks 4

Sol. : Here, $M = x^2 - 4xy - 2y^2$, $N = y^2 - 4xy - 2x^2$

$$\frac{\partial M}{\partial y} = -4x - 4y, \quad \frac{\partial N}{\partial x} = -4y - 4x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is exact.

$\therefore$ Solution is

$$\int M dx + \int N dy = C$$

$$\left(\begin{array}{l} \text{Term of N} \\ \text{free from x} \end{array} \right)$$

$$\Rightarrow \int (x^2 - 4xy - 2y^2) dx + \int y^2 dy = C$$

$$\Rightarrow \frac{x^3}{3} - 2x^2y - 2xy^2 + \frac{y^3}{3} = C$$

$$\Rightarrow \frac{x^3}{3} - 2x^2y - 2xy^2 + \frac{y^3}{3} = C$$

Ex. 2.5.11 : Solve $((e^y + 1) \cos x)dx + (e^y \sin x) dy = 0$.

Sol. : Here, $M = (e^y + 1) \cos x$, $N = e^y \sin x$

$$\frac{\partial M}{\partial y} = e^y \cos x, \quad \frac{\partial N}{\partial x} = e^y \cos x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is exact.

$\therefore$ Solution is

$$\int M dx + \int N dy = C$$

$$\left(\begin{array}{l} \text{Term of N} \\ \text{free from x} \end{array} \right)$$

$$\Rightarrow \int (e^y + 1) \cos x dx + \int 0 dy = C$$

$$\Rightarrow (e^y + 1) \sin x = C$$

2.6 Non-Exact Differential Equation**GTU : Winter-18,19, Summer-18,19,23**

A differential equation of the form $M(x, y) dx + N(x, y) dy = 0$ is called a non-exact differential equation if,

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

To solve non-exact differential equation, we need to multiply it by some function such that it becomes an exact differential equation. The function is called an Integrating Factor (I.F.)

To find an Integrating Factor (I.F.) there are different cases.

Case - I : If $M(x, y) dx + N(x, y) dy = 0$ is a homogeneous differential equation and $Mx + Ny \neq 0$ then integrating factor.

$$\text{I.F.} = \frac{1}{Mx + Ny}$$

Case - II : Let $M(x, y) dx + N(x, y) dy = 0$ be of the form $f_1(x, y)y dx + f_2(x, y)x dy = 0$

If $Mx - Ny \neq 0$ then

$$\text{I.F.} = \frac{1}{Mx - Ny}$$

Case - III : If $\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$ is only function of y i.e. $R(y)$, then

$$\text{I.F.} = e^{\int R(y) dy}$$

Case - IV : If $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is only function of x i.e. $R(x)$, then

$$\text{I.F.} = e^{\int R(x) dx}$$

After multiplying with an integrating factor, the equation becomes exact and it can be solve using formula of exact differential equation.

Ex. 2.6.1 : Solve $x^2 y dx - (x^3 + y^3) dy = 0$.

GTU : Summer-18, Maths-3, Marks 4

Sol. : Here, $M = x^2 y$, $N = -(x^3 + y^3)$

$$\frac{\partial M}{\partial y} = x^2 \quad \frac{\partial N}{\partial x} = -3x^2$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is non-exact.

Given equation is homogeneous differential equation

$$\therefore \text{I.F.} = \frac{1}{Mx + Ny} = \frac{1}{x^3 y - x^3 y - y^4} = \frac{1}{y^4}$$

$$\therefore M^* = \frac{x^2}{y^3}, \quad N^* = -\frac{x^3}{y^4} - \frac{1}{y}$$

$$\frac{\partial M^*}{\partial y} = -\frac{3x^2}{y^4}, \quad \frac{\partial N^*}{\partial x} = -\frac{3x^2}{y^4}$$

$$\therefore \frac{\partial M^*}{\partial y} = \frac{\partial N^*}{\partial x}$$

$\therefore$ Equation is exact for new differential equation.

$\therefore$ Solution is

$$\int M^* dx + \int N^* dy = C$$

$$\left(\begin{array}{l} \text{Term of } N^* \\ \text{free from } x \end{array} \right)$$

$$\Rightarrow \int \frac{x^2}{y^3} dx + \int \left(-\frac{1}{y} \right) dy = C$$

$$\Rightarrow \frac{x^3}{3y^3} - \log y = C$$

Ex. 2.6.2 : Solve $(x^4 + y^4) dx - xy^3 dy = 0$.

GTU : Summer-18, AEM, Marks 3

Sol. : Here, $M = x^4 + y^4$, $N = -xy^3$

$$\frac{\partial M}{\partial y} = 4y^3 \quad \frac{\partial N}{\partial x} = -y^3$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is non-exact.

Given equation is homogeneous differential equation

$$\therefore \text{I.F.} = \frac{1}{Mx + Ny} = \frac{1}{x^5 + xy^4 - xy^4} = \frac{1}{x^5}$$

$$\therefore M^* = \frac{1}{x} + \frac{y^4}{x^5}, \quad N^* = -\frac{y^3}{x^4}$$

$$\frac{\partial M^*}{\partial y} = \frac{4y^3}{x^5}, \quad \frac{\partial N^*}{\partial x} = \frac{4y^3}{x^5}$$

∴ Equation is exact for new differential equation.

∴ Solution is

$$\int M^* dx + \int N^* dy = C$$

(Term of N^*
free from x)

$$\Rightarrow \int \left(\frac{1}{x} + \frac{y^4}{x^5} \right) dx + \int 0 dy = C$$

$$\Rightarrow \log x - \frac{y^4}{4x^4} = C$$

Ex. 2.6.3 : Solve $(x^3 + y^3) dx - xy^2 dy = 0$.

GTU : Summer-23, Maths-2, Marks 3

Sol. : Here, $M = x^3 + y^3$, $N = -xy^2$

$$\frac{\partial M}{\partial y} = 3y^2, \quad \frac{\partial N}{\partial x} = -y^2$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

∴ Given differential equation is non-exact.

Given equation is homogeneous differential equation.

$$\therefore \text{I.F.} = \frac{1}{Mx + Ny} = \frac{1}{x^4 + xy^3 - xy^3} = \frac{1}{x^4}$$

$$\therefore M^* = \frac{1}{x} + \frac{y^3}{x^4}, \quad N^* = -\frac{y^2}{x^3}$$

$$\frac{\partial M^*}{\partial y} = \frac{3y^2}{x^4}, \quad \frac{\partial N^*}{\partial x} = \frac{3y^2}{x^4}$$

∴ Equation is exact for new differential equation.

∴ Solution is

$$\int M^* dx + \int N^* dy = C$$

(Term of N^*
free from x)

$$\Rightarrow \int \left(\frac{1}{x} + \frac{y^3}{x^4} \right) dx + \int 0 dy = C$$

$$\Rightarrow \log x - \frac{y^3}{3x^3} = C$$

Ex. 2.6.4 : Solve $(x^2 - y^2) dx + xy dy = 0$.

GTU : Summer-23, Maths-2, Marks 3

Sol. : Here, $M = x^2 - y^2$, $N = xy$

$$\frac{\partial M}{\partial y} = -2y, \quad \frac{\partial N}{\partial x} = y$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

∴ Given differential equation is non-exact.

Given equation is homogeneous differential equation.

$$\therefore \text{I.F.} = \frac{1}{Mx + Ny} = \frac{1}{x^3 - xy^2 + xy^2} = \frac{1}{x^3}$$

$$\therefore M^* = \frac{1}{x} - \frac{y^2}{x^3}, \quad N^* = \frac{y}{x^2}$$

$$\frac{\partial M^*}{\partial y} = -\frac{2y}{x^3}, \quad \frac{\partial N^*}{\partial x} = -\frac{2y}{x^3}$$

∴ Equation is exact for new differential equation.

∴ Solution is

$$\int M^* dx + \int N^* dy = C$$

(Term of N^*
free from x)

$$\Rightarrow \int \left(\frac{1}{x} - \frac{y^2}{x^3} \right) dx + \int 0 dy = C$$

$$\Rightarrow \log x + \frac{y^2}{2x^2} = C$$

Ex. 2.6.5 : Solve $(x^2 y^2 + 2) y dx + (2 - x^2 y^2) x dy = 0$.

GTU : Winter-18, Maths-3, Marks 4

Sol. : Here, $M = (x^2 y^2 + 2) y$, $N = (2 - x^2 y^2) x$

$$= x^2 y^3 + 2y = 2x - x^3 y^2$$

$$\therefore \frac{\partial M}{\partial y} = 3x^2 y^2 + 2, \quad \frac{\partial N}{\partial x} = 2 - 3x^2 y^2$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is non-exact.

Given equation is of the form

$$f_1(x, y)y \, dx + f_2(x, y)x \, dy = 0$$

$$\therefore \text{I.F.} = \frac{1}{Mx - Ny}$$

$$= \frac{1}{x^3y^3 + 2xy - 2xy + x^3y^3}$$

$$= \frac{1}{2x^3y^3}$$

$$\therefore M^* = \frac{1}{2x} + \frac{1}{x^3y^2}, \quad N^* = \frac{1}{x^2y^3} - \frac{1}{2y}$$

$$\frac{\partial M^*}{\partial y} = -\frac{2}{x^3y^3}, \quad \frac{\partial N^*}{\partial x} = -\frac{2}{x^3y^3}$$

$$\therefore \frac{\partial M^*}{\partial y} = \frac{\partial N^*}{\partial x}$$

$\therefore$ Equation is exact for new differential equation.

$\therefore$ Solution is

$$\int M^* \, dx + \int N^* \, dy = C$$

$$\left(\begin{array}{l} \text{Term of } N^* \\ \text{free from } x \end{array} \right)$$

$$\Rightarrow \int \left(\frac{1}{2x} + \frac{1}{x^3y^2} \right) dx + \int \left(-\frac{1}{2y} \right) dy = C$$

$$\Rightarrow \frac{1}{2} \log x - \frac{1}{2x^2y^2} - \frac{1}{2} \log y = C$$

$$\Rightarrow \frac{1}{2} \log \left(\frac{x}{y} \right) - \frac{1}{2x^2y^2} = C$$

Ex. 2.6.6 : Solve $(x^2y^2 + xy + 1)y \, dx + (x^2y^2 - xy + 1)x \, dy = 0$.

GTU : Winter-19, Maths-3, Marks 3

Sol. : Here,

$$\begin{aligned} M &= (x^2y^2 + xy + 1)y, \quad N = (x^2y^2 - xy + 1)x \\ &= x^2y^3 + xy^2 + y \quad = x^3y^2 - x^2y + x \end{aligned}$$

$$\therefore \frac{\partial M}{\partial y} = 3x^2y^2 + 2xy + 1$$

$$\frac{\partial N}{\partial x} = 3x^2y^2 - 2xy + 1$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is non-exact.

Given equation is of the form

$$f_1(x, y)y \, dx + f_2(x, y)x \, dy = 0$$

$$\therefore \text{I.F.} = \frac{1}{Mx - Ny}$$

$$= \frac{1}{x^3y^3 + x^2y^2 + xy - x^3y^3 + x^2y^2 - xy}$$

$$= \frac{1}{2x^2y^2}$$

$$\therefore M^* = \frac{y}{2} + \frac{1}{2x} + \frac{1}{2x^2y}, \quad N^* = \frac{x}{2} - \frac{1}{2y} + \frac{1}{2xy^2}$$

$$\frac{\partial M^*}{\partial y} = \frac{1}{2} - \frac{1}{2x^2y^2}, \quad \frac{\partial N^*}{\partial x} = \frac{1}{2} - \frac{1}{2x^2y^2}$$

$\therefore$ Equation is exact for new differential equation.

$\therefore$ Solution is

$$\int M^* \, dx + \int N^* \, dy = C$$

$$\left(\begin{array}{l} \text{Term of } N^* \\ \text{free from } x \end{array} \right)$$

$$\Rightarrow \int \left(\frac{y}{2} + \frac{1}{2x} + \frac{1}{2x^2y} \right) dx + \int \left(-\frac{1}{2y} \right) dy = C$$

$$\Rightarrow \frac{xy}{2} + \frac{1}{2} \log x - \frac{1}{2xy} - \frac{1}{2} \log y = C$$

$$\Rightarrow \frac{xy}{2} + \frac{1}{2} \log \left(\frac{x}{y} \right) - \frac{1}{2xy} = C$$

Ex. 2.6.7 : Solve $(xy + 1)y \, dx + (2 - xy)x \, dy = 0$.

Sol. : Here, $M = xy^2 + y$, $N = 2x - x^2y$

$$\frac{\partial M}{\partial y} = 2xy + 1, \quad \frac{\partial N}{\partial x} = 2 - 2xy$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

∴ Given differential equation is non-exact.

Given equation is of the form

$$f_1(x, y)y dx + f_2(x, y)x dy = 0$$

$$\begin{aligned} \therefore \text{I.F.} &= \frac{1}{Mx - Ny} = \frac{1}{x^2y^2 + xy - 2xy + x^2y^2} \\ &= \frac{1}{2x^2y^2 - xy} = \frac{1}{xy(2xy - 1)} \end{aligned}$$

$$\therefore M^* = \frac{xy+1}{x(2xy-1)}, \quad N^* = \frac{2-xy}{y(2xy-1)}$$

Equation becomes exact for M^* and N^* .

Using partial fraction,

$$\frac{xy+1}{x(2xy-1)} = \frac{A}{x} + \frac{B}{2xy-1}$$

$$\Rightarrow xy + 1 = A(2xy - 1) + Bx$$

$$\Rightarrow xy + 1 = (2Ay + B)x - A$$

$$\therefore A = -1, \quad 2yA + B = y$$

$$\Rightarrow B = 3y$$

$$\therefore \frac{xy+1}{x(2xy-1)} = -\frac{1}{x} + \frac{3y}{2xy-1}$$

Again, using partial fraction,

$$\frac{2-xy}{y(2xy-1)} = \frac{A}{y} + \frac{B}{2xy-1}$$

$$\Rightarrow 2 - xy = A(2xy - 1) + By$$

$$\Rightarrow 2 - xy = (2xA + B)y - A$$

$$\therefore A = -2, \quad 2Ax + B = -x$$

$$\Rightarrow B = 3x$$

$$\therefore \frac{2-xy}{y(2xy-1)} = -\frac{2}{y} + \frac{3x}{2xy-1}$$

Solution is

$$\int M^* dx + \int N^* dy = C$$

$$\left(\begin{array}{l} \text{Term of } N^* \\ \text{free from } x \end{array} \right)$$

$$\Rightarrow \int \left(-\frac{1}{x} + \frac{3y}{2xy-1} \right) dx + \int \left(-\frac{2}{y} \right) dy = C$$

$$\Rightarrow -\log x + \frac{3}{2} \log(2xy-1) - 2 \log y = C$$

$$\Rightarrow -2 \log x + 3 \log(2xy-1) - 4 \log y = C$$

$$\Rightarrow \frac{(2xy-1)^3}{x^2y^4} = C$$

Ex. 2.6.8 : Solve $2xy dx + (3x^2 + y) dy = 0$.

Sol. : Here, $M = 2xy$, $N = 3x^2 + y$

$$\frac{\partial M}{\partial y} = 2x, \quad \frac{\partial N}{\partial x} = 6x$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

∴ Given differential equation is non-exact.

$$\begin{aligned} \text{Let } R &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{2xy} (6x - 2x) = \frac{4x}{2xy} = \frac{2}{y} = R(y) \end{aligned}$$

$$\begin{aligned} \therefore \text{I.F.} &= e^{\int R(y) dy} \\ &= e^{\int \frac{2}{y} dy} = e^{2 \log y} = y^2 \end{aligned}$$

$$\therefore M^* = 2xy^3, \quad N^* = 3x^2y^2 + y^3$$

$$\frac{\partial M^*}{\partial y} = 6xy^2, \quad \frac{\partial N^*}{\partial x} = 6xy^2$$

∴ Equation is exact for new differential equation.

∴ Solution is

$$\int M^* dx + \int N^* dy = C$$

$$\left(\begin{array}{l} \text{Term of } N^* \\ \text{free from } x \end{array} \right)$$

$$\Rightarrow \int 2xy^3 dx + \int y^3 dy = C$$

$$\Rightarrow x^2y^3 + \frac{y^4}{4} = C$$

Ex. 2.6.9 : Solve $(x^2 - y^2) dy = 2xy dx$.

GTU : Summer-18, AEM, Marks 3

$$\text{Sol. : } (x^2 - y^2) dy = 2xy \cdot dx$$

$$\Rightarrow 2xy dx + (y^2 - x^2) dy = 0$$

$$\text{Here, } M = 2xy, \quad N = y^2 - x^2$$

$$\frac{\partial M}{\partial y} = 2x, \quad \frac{\partial N}{\partial x} = -2x$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is non-exact.

$$\begin{aligned} \text{Let } R &= \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ &= \frac{1}{2xy} (-2x - 2x) \\ &= \frac{-4x}{2xy} = \frac{-2}{y} = R(y) \end{aligned}$$

$$\begin{aligned} \therefore \text{I.F.} &= e^{\int R(y) dy} = e^{\int \frac{-2}{y} dy} \\ &= e^{-2 \log y} = \frac{1}{y^2} \end{aligned}$$

$$\therefore M^* = \frac{2x}{y}, \quad N^* = 1 - \frac{x^2}{y^2}$$

$$\frac{\partial M^*}{\partial y} = -\frac{2x}{y^2}, \quad \frac{\partial N^*}{\partial x} = -\frac{2x}{y^2}$$

$\therefore$ Equation is exact for new differential equation.

$\therefore$ Solution is

$$\begin{aligned} \int M^* dx + \int N^* dy &= C \\ &\left(\begin{array}{l} \text{Term of } N^* \\ \text{free from } x \end{array} \right) \\ \Rightarrow \int \frac{2x}{y} dx + \int (1) dy &= C \end{aligned}$$

$$\Rightarrow \frac{x^2}{y} + y = C$$

Ex. 2.6.10 : Solve $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$.

Sol. : Here, $M = xy^3 + y$, $N = 2(x^2y^2 + x + y^4)$

$$\frac{\partial M}{\partial y} = 3xy^2 + 1, \quad \frac{\partial N}{\partial x} = 4xy^2 + 2$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is non-exact.

$$\text{Let } R = \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$$

$$= \frac{4xy^2 + 2 - 3xy^2 - 1}{(xy^2 + 1)y}$$

$$= \frac{xy^2 + 1}{(xy^2 + 1)y} = \frac{1}{y} = R(y)$$

$$\begin{aligned} \text{I.F.} &= e^{\int R(y) dy} = e^{\int \frac{1}{y} dy} \\ &= e^{\log y} = y \end{aligned}$$

$$\therefore M^* = xy^4 + y^2, \quad N^* = 2(x^2y^3 + xy + y^3)$$

$$\frac{\partial M^*}{\partial y} = 4xy^3 + 2y, \quad \frac{\partial N^*}{\partial x} = 4xy^3 + 2y$$

$\therefore$ Equation is exact for new differential equation.

$\therefore$ Solution is

$$\begin{aligned} \int M^* dx + \int N^* dy &= C \\ &\left(\begin{array}{l} \text{Term of } N^* \\ \text{free from } x \end{array} \right) \end{aligned}$$

$$\Rightarrow \int (xy^4 + y^2) dx + \int 2y^5 dy = C$$

$$\Rightarrow \frac{x^2y^4}{2} + xy^2 + \frac{y^6}{3} = C$$

Ex. 2.6.11 : Solve $(2x^3 + 4y)dx - x dy = 0$.

GTU : Summer-19, Maths-2, Marks 3

Sol. : Here, $M = 2x^3 + 4y$, $N = -x$

$$\frac{\partial M}{\partial y} = 4, \quad \frac{\partial N}{\partial x} = -1$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ Given differential equation is non-exact.

$$\begin{aligned} \text{Let } R &= \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{-x} (4 + 1) \\ &= -\frac{5}{x} = R(x) \end{aligned}$$

$$\begin{aligned} \text{I.F.} &= e^{\int R(x) dx} = e^{\int -\frac{5}{x} dx} = e^{-5 \log x} \\ &= x^{-5} = \frac{1}{x^5} \end{aligned}$$

$$\therefore M^* = \frac{2}{x^2} + \frac{4y}{x^5}, \quad N^* = -\frac{1}{x^4}$$

$$\frac{\partial M^*}{\partial y} = \frac{4}{x^5}, \quad \frac{\partial N^*}{\partial x} = \frac{4}{x^5}$$

∴ Equation is exact for new differential equation.

∴ Solution is

$$\int M^* dx + \int N^* dy = C$$

$$\left(\begin{array}{c} \text{Term of } N^* \\ \text{free from } x \end{array} \right)$$

$$\Rightarrow \int \left(\frac{2}{x^2} + \frac{4y}{x^5} \right) dx + \int 0 dy = C$$

$$\Rightarrow -\frac{2}{x} - \frac{y}{x^4} = C$$

$$\Rightarrow 2x^3 + y = Cx^4$$

Ex. 2.6.12 : Solve $3e^x \tan y dx + (1+e^x) \sec^2 y dy = 0$.

GTU : Summer-23, AEM, Marks 3

Sol. : Here, $M = 3e^x \tan y$, $N = (1+e^x) \sec^2 y$

$$\frac{\partial M}{\partial y} = 3e^x \sec^2 y, \quad \frac{\partial N}{\partial x} = e^x \sec^2 y$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

∴ Given differential equation is non-exact.

Let $R = \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$

$$= \frac{3e^x \sec^2 y - e^x \sec^2 y}{(1+e^x) \sec^2 y}$$

$$= \frac{2e^x \sec^2 y}{(1+e^x) \sec^2 y} = \frac{2e^x}{1+e^x} = R(x).$$

$$\text{I.F.} = e^{\int R(x) dx}$$

$$= e^{\int \frac{2e^x}{1+e^x} dx}$$

$$= e^{2 \log(1+e^x)} = (1+e^x)^2$$

$$\therefore M^* = 3e^x(1+e^x)^2 \tan y$$

$$N^* = (1+e^x)^3 \sec^2 y$$

$$\frac{\partial M^*}{\partial y} = 3e^x(1+e^x)^2 \sec^2 y$$

$$\frac{\partial N^*}{\partial x} = 3e^x(1+e^x)^2 \sec^2 y$$

∴ Equation is exact for new differential equation.

∴ Solution is

$$\int M^* dx + \int N^* dy = C$$

$$\left(\begin{array}{c} \text{Term of } N^* \\ \text{free from } x \end{array} \right)$$

$$\Rightarrow \int 3e^x(1+e^x)^2 \tan y dx + \int \sec^2 y dy = C$$

$$\Rightarrow (1+e^x)^3 \tan y + \tan y = C$$

Ex. 2.6.13 : Solve $(x+2y^2) dx + 2xy dy = 0$.

Sol. : Here, $M = x+2y^2$, $N = 2xy$

$$\frac{\partial M}{\partial y} = 4y, \quad \frac{\partial N}{\partial x} = 2y$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

∴ Given differential equation is non-exact.

Let $R = \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$

$$= \frac{4y - 2y}{2xy} = \frac{2y}{2xy} = \frac{1}{x} = R(x)$$

$$\text{I.F.} = e^{\int R(x) dx} = e^{\int \frac{1}{x} dx} = e^{\log x}$$

$$= x$$

$$\therefore M^* = x^2 + 2xy^2, \quad N^* = 2x^2y$$

$$\frac{\partial M^*}{\partial y} = 4xy, \quad \frac{\partial N^*}{\partial x} = 4xy$$

∴ Equation is exact for new differential equation.

∴ Solution is

$$\int M^* dx + \int N^* dy = C$$

$$\left(\begin{array}{c} \text{Term of } N^* \\ \text{free from } x \end{array} \right)$$

$$\Rightarrow \int (x^2 + 2xy^2) dx + \int 0 dy = C$$

$$\Rightarrow \frac{x^3}{3} + x^2y^2 = C$$

2.7 Linear Differential Equations**GTU : Winter-19,22,24, Summer-18,19,20,22,23,24**

A differential equation of first degree and not multiplied with dependent variable is called a first order linear differential equation.

The standard form of a first order linear differential equation is,

$\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x)$ and $Q(x)$ are functions of only x .

Solution of linear equation is

$$y \cdot (\text{I.F.}) = \int Q(x) \cdot \text{I.F.} dx + C$$

where, $\text{I.F.} = e^{\int P(x) dx}$

I.F. is called integrating factor.

Ex. 2.7.1 : Solve $\frac{dy}{dx} - 2y = 4 - x$.

GTU : Summer-22, Maths-2, Marks 3

Sol. :

Comparing with $\frac{dy}{dx} + P(x)y = Q(x)$

$\therefore P(x) = -2, Q(x) = 4 - x$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x) dx} \\ &= e^{\int -2 dx} = e^{-2x} \end{aligned}$$

$\therefore$ Solution is

$$y \cdot (\text{I.F.}) = \int Q(x) (\text{I.F.}) dx + C$$

$$y(e^{-2x}) = \int (4-x)e^{-2x} dx + C$$

$$\Rightarrow y \cdot e^{-2x} = (4-x) \int e^{-2x} dx - \int \left(\frac{d}{dx} (4-x) \int e^{-2x} dx \right) dx + C$$

$$\Rightarrow y \cdot e^{-2x} = (4-x) \left(\frac{e^{-2x}}{-2} \right) - \int (-1) \left(\frac{e^{-2x}}{-2} \right) dx + C$$

$$\Rightarrow ye^{-2x} = \frac{4-x}{(-2)} e^{-2x} - \int \frac{e^{-2x}}{2} dx + C$$

$$\Rightarrow ye^{-2x} = \frac{x-4}{2} e^{-2x} - \left(\frac{e^{-2x}}{-4} \right) + C$$

$$\Rightarrow ye^{-2x} = \frac{x-4}{2} e^{-2x} + \frac{e^{-2x}}{4} + C$$

$$\Rightarrow y = \frac{x-4}{2} + \frac{1}{4} + Ce^{2x}$$

$$\Rightarrow y = \frac{x}{4} - \frac{7}{4} + Ce^{2x}$$

Ex. 2.7.2 : Solve $\frac{dy}{dx} + \frac{4xy}{x^2+1} = \frac{1}{(x^2+1)^3}$

GTU : Summer-23, AEM, Marks

Sol. : Here, $P(x) = \frac{4x}{x^2+1}, Q(x) = \frac{1}{(x^2+1)^3}$

$$\text{I.F.} = e^{\int P(x) dx}$$

$$= e^{\int \frac{4x}{x^2+1} dx}$$

$$= e^{2 \log(x^2+1)} = (x^2+1)^2$$

$\therefore$ Solution is

$$y \cdot (\text{I.F.}) = \int Q(x) (\text{I.F.}) dx + C$$

$$\Rightarrow y(x^2+1)^2 = \int \frac{1}{(x^2+1)^3} (x^2+1)^2 dx + C$$

$$\Rightarrow y(x^2+1)^2 = \int \frac{1}{x^2+1} dx + C$$

$$\Rightarrow y(x^2+1)^2 = \tan^{-1} x + C$$

Ex. 2.7.3 : Solve $\frac{dy}{dx} + y = e^{-x}, y(1) = 1$.

GTU : Winter-24, Maths-2, Marks

Sol. : Here, $P(x) = 1, Q(x) = e^{-x}$

$$\text{I.F.} = e^{\int P(x) dx} = e^{\int 1 dx} = e^x$$

$\therefore$ Solution is

$$y \cdot (\text{I.F.}) = \int Q(x) (\text{I.F.}) dx + C$$

$$\Rightarrow ye^x = \int e^{-x} \cdot e^x dx + C$$

$$\Rightarrow ye^x = \int 1 dx + C$$

$$\Rightarrow ye^x = x + C$$

$$\Rightarrow y = (x+C)e^{-x}$$

which is general solution.

Now, $y(1) = 1$

$$\Rightarrow 1 = (1+C)e^{-1}$$

$$\Rightarrow e = 1 + C \Rightarrow C = e - 1$$

$\therefore$ The particular solution is

$$y = (x+e-1)e^{-x}$$

Ex. 2.7.4 : Solve $\frac{dy}{dx} - y \cot x = 2x \sin x$.

GTU : Winter-22, Maths-2, Marks 3

Sol. : Here, $P(x) = -\cot x$, $Q(x) = 2x \sin x$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x)dx} \\ &= e^{\int -\cot x dx} \\ &= e^{-\log(\sin x)} \\ &= (\sin x)^{-1} \end{aligned}$$

Solution is

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int Q(x)(\text{I.F.})dx + C \\ \Rightarrow y(\sin x)^{-1} &= \int 2x \sin x (\sin x)^{-1} dx + C \\ \Rightarrow y(\sin x)^{-1} &= \int 2x dx + C \\ \Rightarrow y(\sin x)^{-1} &= x^2 + C \\ \Rightarrow y &= (x^2 + C) \sin x \end{aligned}$$

Ex. 2.7.5 : Solve $\frac{dy}{dx} + y \cot x = \cos x$.

GTU : Summer-19, Maths-3, Summer-20, Maths-2 Marks 4

Sol. : Here, $P(x) = \cot x$, $Q(x) = \cos x$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x)dx} = e^{\int \cot x dx} \\ &= e^{\log(\sin x)} \\ &= \sin x \end{aligned}$$

Solution is

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int Q(x)(\text{I.F.})dx + C \\ \Rightarrow y(\sin x) &= \int \cos x \cdot (\sin x) dx + C \\ \Rightarrow y \cdot \sin x &= \int \frac{1}{2} \sin 2x dx + C \\ \Rightarrow y \cdot \sin x &= -\frac{\cos 2x}{4} + C \\ \Rightarrow 4y \sin x &= C - \cos 2x \end{aligned}$$

Ex. 2.7.6 : Solve $y' + \frac{1}{x^2} y = e^{\frac{1}{x}}$.

GTU : Summer-20, Maths-3, Marks 3

Sol. : Here, $P(x) = \frac{1}{x^2}$, $Q(x) = e^{\frac{1}{x}}$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x)dx} = e^{\int \frac{1}{x^2} dx} \\ &= e^{-\frac{1}{x}} e^{\int P(x)dx} \end{aligned}$$

Solution is

$$\begin{aligned} y(\text{I.F.}) &= \int Q(x)(\text{I.F.})dx + C \\ \Rightarrow y \cdot e^{-\frac{1}{x}} &= \int e^{\frac{1}{x}} \cdot e^{-\frac{1}{x}} dx + C \\ \Rightarrow y \cdot e^{-\frac{1}{x}} &= \int 1 dx + C \\ \Rightarrow y \cdot e^{-\frac{1}{x}} &= x + C \\ \Rightarrow y &= (x + C)e^{\frac{1}{x}} \end{aligned}$$

Ex. 2.7.7 : Solve $\frac{dy}{dx} + y \sin x = e^{\cos x}$.

GTU : Summer-18, AEM, Summer-22, 23, 24, Winter-22, Maths-3, Marks 3

Sol. : Here, $P(x) = \sin x$, $Q(x) = e^{\cos x}$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x)dx} = e^{\int \sin x dx} \\ &= e^{-\cos x} \end{aligned}$$

Solution is

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int Q(x)(\text{I.F.})dx + C \\ \Rightarrow y \cdot e^{-\cos x} &= \int e^{\cos x} \cdot e^{-\cos x} dx + C \\ \Rightarrow y \cdot e^{-\cos x} &= \int 1 dx + C \\ \Rightarrow y \cdot e^{-\cos x} &= x + C \\ \Rightarrow y &= (x + C)e^{\cos x} \end{aligned}$$

Ex. 2.7.8 : Solve $\frac{dy}{dx} + 2xy = 2e^{-x^2}$.

GTU : Winter-19, Maths-3, Summer-20, AEM, Marks 3

Sol. : Here, $P(x) = 2x$, $Q(x) = 2e^{-x^2}$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x)dx} = e^{\int 2x dx} \\ &= e^{x^2} \end{aligned}$$

Solution is

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int Q(x)(\text{I.F.})dx + C \\ \Rightarrow y \cdot e^{x^2} &= \int 2e^{-x^2} \cdot e^{x^2} dx + C \end{aligned}$$

$$\Rightarrow ye^{x^2} = \int 2x dx + C$$

$$\Rightarrow ye^{x^2} = 2x + C$$

$$\Rightarrow y = (2x + C)e^{-x^2}$$

Ex. 2.7.9 : Solve $\frac{dy}{dx} + 2y \tan x = \sin x$

GTU : Winter-19, 21, AEM, Winter-23, Maths-3, Summer-23, Maths-2, Marks 3

Sol. : Here, $P(x) = 2 \tan x$, $Q(x) = \sin x$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x) dx} = e^{\int 2 \tan x dx} = e^{2 \log (\sec x)} \\ &= \sec^2 x \end{aligned}$$

Solution is

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int Q(x) \cdot (\text{I.F.}) dx + C \\ \Rightarrow y(\sec^2 x) &= \int \sin x \cdot \sec^2 x dx + C \\ \Rightarrow y(\sec^2 x) &= \int \tan x \cdot \sec x dx + C \\ \Rightarrow y(\sec^2 x) &= \sec x + C \\ \Rightarrow y &= \cos x + C \cdot \cos^2 x \end{aligned}$$

Ex. 2.7.10 : Solve $(x+1) \frac{dy}{dx} - y = e^{3x} (x+1)^2$.

GTU : Summer-19, Maths-2, Marks 4

Sol. : $(x+1) \frac{dy}{dx} - y = e^{3x} (x+1)^2$

$$\Rightarrow \frac{dy}{dx} - \frac{1}{x+1} y = e^{3x} (x+1)$$

$$\therefore P(x) = -\frac{1}{x+1}, Q(x) = e^{3x} (x+1)$$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x) dx} = e^{\int -\frac{1}{x+1} dx} \\ &= e^{-\log (x+1)} \\ &= \frac{1}{x+1} \end{aligned}$$

Solution is

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int Q(x) \cdot (\text{I.F.}) dx + C \\ \Rightarrow y \cdot \frac{1}{x+1} &= \int e^{3x} (x+1) \frac{1}{x+1} dx + C \\ \Rightarrow y \cdot \frac{1}{x+1} &= \int e^{3x} dx + C \end{aligned}$$

$$\Rightarrow y \frac{1}{x+1} = \frac{e^{3x}}{3} + C$$

$$\Rightarrow y = \left(\frac{e^{3x}}{3} + C \right) (x+1)$$

Ex. 2.7.11 : Solve $y' - (1+3x^{-1})y = x+2$, $y(1) = e-1$

GTU : Summer-24, AEM, Marks 4

Sol. : Here, $P(x) = -(1+3x^{-1})$, $Q(x) = x+2$

$$= -1 - \frac{3}{x}$$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x) dx} = e^{\int \left(-1 - \frac{3}{x}\right) dx} \\ &= e^{-x - 3 \log x} \\ &= e^{-x} \cdot e^{-3 \log x} \\ &= e^{-x} x^{-3} \end{aligned}$$

Solution is

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int Q(x) \cdot (\text{I.F.}) dx + C \\ \Rightarrow ye^{-x} x^{-3} &= \int (x+2) e^{-x} x^{-3} dx + C \\ \Rightarrow ye^{-x} x^{-3} &= \int \left(\frac{1}{x^2} + \frac{2}{x^3} \right) e^{-x} dx + C \\ \Rightarrow ye^{-x} x^{-3} &= \int \frac{1}{x^2} e^{-x} dx + \int \frac{2}{x^3} e^{-x} dx + C \\ \Rightarrow ye^{-x} x^{-3} &= \frac{1}{x^2} (-e^{-x}) - \int \left(-\frac{2}{x^3} \right) (-e^{-x}) dx \\ &\quad + \int \frac{2}{x^3} e^{-x} dx + C \\ \Rightarrow ye^{-x} x^{-3} &= -\frac{e^{-x}}{x^2} - \int \frac{2}{x^3} e^{-x} dx + \int \frac{2}{x^3} e^{-x} dx + C \\ \Rightarrow ye^{-x} x^{-3} &= -\frac{e^{-x}}{x^2} + C \\ \Rightarrow y &= -x + Ce^x x^3 \end{aligned}$$

which is general solution.

Now, $y(1) = e - 1$

$$\Rightarrow e - 1 = -1 + Ce$$

$$\Rightarrow C = 1$$

$\therefore$ The particular solution is

$$y = e^x x^3 - x$$

Ex. 2.7.12 : Solve $(1+y^2)dx + (x - e^{-\tan^{-1}y})dy = 0$.

**GTU : Winter-19, Maths-3, Marks 4,
Summer-19, AEM, Summer-19, Maths-2**

Sol. : $(1+y^2)dx + (x - e^{-\tan^{-1}y})dy = 0$

$$\Rightarrow (1+y^2)dx = (e^{-\tan^{-1}y} - x)dy$$

$$\Rightarrow (1+y^2)\frac{dx}{dy} = e^{-\tan^{-1}y} - x$$

$$\Rightarrow \frac{dx}{dy} + \frac{1}{1+y^2}x = \frac{e^{-\tan^{-1}y}}{1+y^2}$$

$$\therefore \text{Here, } P(y) = \frac{1}{1+y^2}, Q(y) = \frac{e^{-\tan^{-1}y}}{1+y^2}$$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x)dy} = e^{\int \frac{1}{1+y^2} dy} \\ &= e^{\tan^{-1}y} \end{aligned}$$

$\therefore$ Solution is

$$x \cdot (\text{I.F.}) = \int Q(y)(\text{I.F.})dy + C$$

$$\Rightarrow x e^{\tan^{-1}y} = \int \frac{e^{-\tan^{-1}y}}{1+y^2} \cdot e^{\tan^{-1}y} dy + C$$

$$\Rightarrow x \cdot e^{\tan^{-1}y} = \int \frac{1}{1+y^2} dy + C$$

$$\Rightarrow x \cdot e^{\tan^{-1}y} = \tan^{-1}y + C$$

$$\Rightarrow x = (\tan^{-1}y + C)e^{-\tan^{-1}y}$$

Ex. 2.7.13 : Solve $\frac{dy}{dx} + \tan x \cdot y = \sin 2x$.

Sol. : Here, $P(x) = \tan x$, $Q(x) = \sin 2x$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x)dx} = e^{\int \tan x \cdot dx} \\ &= e^{\log \sec x} = \sec x \end{aligned}$$

Solution is

$$y \cdot (\text{I.F.}) = \int Q(x)(\text{I.F.})dx + C$$

$$\Rightarrow y \cdot \sec x = \int \sin 2x \cdot \sec x dx + C$$

$$\Rightarrow y \cdot \sec x = \int 2 \sin x dx + C$$

$$\Rightarrow y \cdot \sec x = -2 \cos x + C$$

$$\Rightarrow y = C \cdot \cos x - 2 \cos^2 x$$

Ex. 2.7.14 : Solve $\frac{dy}{dx} = \frac{1}{e^{2y} + x}$.

Sol. : $\frac{dy}{dx} = \frac{1}{e^{2y} + x}$

$$\Rightarrow \frac{dx}{dy} = e^{2y} + x$$

$$\Rightarrow \frac{dx}{dy} - x = e^{2y}$$

$$\therefore P(y) = -1, Q(y) = e^{2y}$$

$$\text{I.F.} = e^{\int P(y)dy} = e^{\int -1 dy} = e^{-y}$$

Solution is

$$x \cdot (\text{I.F.}) = \int Q(y)(\text{I.F.})dy + C$$

$$\Rightarrow x e^{-y} = \int e^{2y} \cdot e^{-y} dy + C$$

$$\Rightarrow x e^{-y} = \int e^y dy + C$$

$$\Rightarrow x e^{-y} = e^y + C$$

$$\Rightarrow x = e^{2y} + C e^y$$

2.8 Bernoulli's Equation

GTU : Winter-19,21,22,23, Summer-19,20

A first order differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^a \text{ is called Bernoulli's}$$

equation, where a is a real number ($a \neq 0, 1$)

The equation can also be written as

$$y^{-a} \frac{dy}{dx} + P(y)y^{1-a} = Q(x)$$

Bernoulli's equation can be solved by reducing into linear equation by substitution.

$$u = y^{1-a}$$

Then using formula of solution of linear equation, evaluate solution of it.

Ex. 2.8.1 : Solve $\frac{dy}{dx} + \frac{y}{x} = y^3$.

GTU : Winter-22, Maths-2, Marks 3

Sol. : Given equation is Bernoulli's equation with $a = 3$.

$$\therefore \text{Let } u = y^{1-a} = y^{1-3} = y^{-2}$$

$$\frac{du}{dx} = -2y^{-3} \frac{dy}{dx}$$

$$\Rightarrow y^{-3} \frac{dy}{dx} = -\frac{1}{2} \frac{du}{dx}$$

Divide given equation by y^3 on both sides,

$$\therefore y^{-3} \frac{dy}{dx} + \frac{1}{x} y^{-2} = 1$$

$$\therefore -\frac{1}{2} \frac{du}{dx} + \frac{1}{x} u = 1$$

$$\Rightarrow \frac{du}{dx} - \frac{2}{x} u = -2$$

$$\therefore P(x) = -\frac{2}{x}, Q(x) = -2$$

$$\text{I.F.} = e^{\int P(x) dx} = e^{\int -\frac{2}{x} dx}$$

$$= e^{-2 \log x}$$

$$= x^{-2}$$

$\therefore$ Solution is

$$u \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.}) dx + C$$

$$\Rightarrow u \cdot x^{-2} = \int \left(-\frac{1}{2}\right) x^{-2} dx + C$$

$$\Rightarrow u \cdot x^{-2} = -\frac{1}{2} \frac{x^{-3}}{-3} + C$$

$$\Rightarrow u x^{-2} = \frac{x^{-3}}{6} + C$$

$$\Rightarrow u = \frac{1}{6x} + C$$

$$\Rightarrow y^{-2} = \frac{1}{6x} + C$$

Ex. 2.8.2 : Solve $\frac{dy}{dx} + \frac{y}{x} - \sqrt{y} = 0$.

GTU : Summer-19, Maths-2, Marks 3

$$\text{Sol. : } \frac{dy}{dx} + \frac{y}{x} = \sqrt{y}$$

$\therefore$ Given equation is Bernoulli's equation with $a = \frac{1}{2}$

$$\text{Let } u = y^{1-a}$$

$$= y^{1-\frac{1}{2}} = y^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2\sqrt{y}} \frac{dy}{dx}$$

$$\Rightarrow \frac{1}{\sqrt{y}} \frac{dy}{dx} = 2 \frac{du}{dx}$$

Divide given equation by $\sqrt{y}$ on both sides,

$$\therefore \frac{1}{\sqrt{y}} \frac{dy}{dx} + \frac{\sqrt{y}}{x} = 1$$

$$\Rightarrow 2 \frac{du}{dx} + \frac{u}{x} = 1$$

$$\Rightarrow \frac{du}{dx} + \frac{1}{2x} u = \frac{1}{2}$$

$$\therefore P(x) = \frac{1}{2x}, Q(x) = \frac{1}{2}$$

$$\text{I.F.} = e^{\int P(x) dx} = e^{\int \frac{1}{2x} dx}$$

$$= e^{\frac{1}{2} \log x}$$

$$= x^{\frac{1}{2}} = \sqrt{x}$$

$\therefore$ Solution is

$$u \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.}) dx + C$$

$$\Rightarrow u \cdot \sqrt{x} = \int \frac{1}{2} \sqrt{x} dx + C$$

$$\Rightarrow u \cdot \sqrt{x} = \frac{1}{2} \frac{x^{3/2}}{3/2} + C$$

$$\Rightarrow u \cdot \sqrt{x} = \frac{1}{3} x^{3/2} + C$$

$$\Rightarrow \sqrt{y} \cdot \sqrt{x} = \frac{1}{3} x^{3/2} + C$$

$$\Rightarrow \sqrt{xy} = \frac{1}{3} x^{3/2} + C$$

Ex. 2.8.3 : Solve $x^2 \frac{dy}{dx} + xy = x^4 y^6$.

GTU : Winter-22, AEM, Marks 4

$$\text{Sol. : } \frac{dy}{dx} + \frac{y}{x} = x^2 y^6$$

It is Bernoulli's equation with $a = 6$.

$$\text{Let } u = y^{1-a} = y^{1-6} = y^{-5}$$

$$\frac{du}{dx} = -5y^{-6} \frac{dy}{dx}$$

$$\Rightarrow y^{-6} \frac{dy}{dx} = -\frac{1}{5} \frac{du}{dx}$$

Divide given equation by y^6 on both sides,

$$\therefore y^{-6} \frac{dy}{dx} + \frac{y^{-5}}{x} = x^2$$

$$\Rightarrow -\frac{1}{5} \frac{du}{dx} + \frac{u}{x} = x^2$$

$$\Rightarrow \frac{du}{dx} - \frac{5}{x} u = -5x^2$$

$$\text{Here, } P(x) = -\frac{5}{x}, \quad Q(x) = -5x^2$$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x) dx} = e^{\int -\frac{5}{x} dx} = e^{-5 \log x} \\ &= x^{-5} \end{aligned}$$

$\therefore$ Solution is

$$u \cdot (\text{I.F.}) = \int Q(x)(\text{I.F.}) dx + C$$

$$\Rightarrow u \cdot x^{-5} = \int (-5x^2) x^{-5} dx + C$$

$$\Rightarrow u x^{-5} = -5 \int x^{-3} dx + C$$

$$\Rightarrow u x^{-5} = -5 \left(\frac{x^{-2}}{-2} \right) + C$$

$$\Rightarrow u x^{-5} = \frac{5}{2} x^{-2} + C$$

$$\Rightarrow u = \frac{5}{2} x^3 + C x^5$$

$$\Rightarrow y^{-5} = \frac{5}{2} x^3 + C x^5$$

Non-linear equation reducible into linear equation :

The equation is of the form

$$f'(y) \frac{dy}{dx} + P(x)f(y) = Q(x)$$

is non-linear first order differential equation.

Equation can be reduced to linear differential equation by substitution.

$$u = f(y)$$

Then using formula of solution of linear equation evaluate solution of it.

$$\text{Ex. 2.8.4 : Solve } \frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2}.$$

GTU : Summer-19, Winter-23, Maths-2, Winter-21, Maths-3, Marks 5

$$\text{Sol. : } \frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2}$$

Divide both sides by e^y

$$e^{-y} \frac{dy}{dx} + e^{-y} \frac{1}{x} = \frac{1}{x^2}$$

$$\text{Let } u = e^{-y}$$

$$\Rightarrow \frac{du}{dx} = -e^{-y} \frac{dy}{dx}$$

$$\Rightarrow e^{-y} \frac{dy}{dx} = -\frac{du}{dx}$$

$$\therefore -\frac{du}{dx} + \frac{1}{x} u = \frac{1}{x^2}$$

$$\Rightarrow \frac{du}{dx} - \frac{1}{x} u = -\frac{1}{x^2}$$

$$\text{Here, } P(x) = -\frac{1}{x} \quad Q(x) = -\frac{1}{x^2}$$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x) dx} = e^{\int -\frac{1}{x} dx} = e^{-\log x} \\ &= x^{-1} = \frac{1}{x} \end{aligned}$$

$\therefore$ Solution is

$$u \cdot (\text{I.F.}) = \int Q(x)(\text{I.F.}) dx + C$$

$$\Rightarrow u \cdot \frac{1}{x} = \int \left(-\frac{1}{x^2} \right) \left(\frac{1}{x} \right) dx + C$$

$$\Rightarrow \frac{u}{x} = -\int \frac{1}{x^3} dx + C$$

$$\Rightarrow \frac{u}{x} = \frac{x^{-2}}{2} + C$$

$$\Rightarrow \frac{e^{-y}}{x} = \frac{1}{2x^2} + C$$

$$\text{Ex. 2.8.5 : Solve } \sec^2 y \frac{dy}{dx} + x \tan y = x^3.$$

GTU : Winter-19, Maths-2, Marks 4

Sol. : Let $u = \tan y$

$$\Rightarrow \frac{du}{dx} = \sec^2 y \frac{dy}{dx}$$

$$\therefore \frac{du}{dx} + xu = x^3$$

$$\text{Here, } P(x) = x, \quad Q(x) = x^3$$

$$\text{I.F.} = e^{\int P(x) dx} = e^{\int x dx} = e^{x^2/2}$$

Solution is

$$u \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.}) dx + C$$

$$\Rightarrow u \cdot e^{x^2/2} = \int x^3 \cdot e^{x^2/2} dx + C$$

$$\text{Let } x^2/2 = t$$

$$x dx = dt$$

$$\therefore u \cdot e^{x^2/2} = \int x^2 \cdot e^{x^2/2} x dx + C$$

$$\Rightarrow u e^{x^2/2} = \int 2t \cdot e^t dt + C$$

$$\Rightarrow u e^{x^2/2} = 2te^t - \int 2e^t dt + C$$

$$\Rightarrow u e^{x^2/2} = 2te^t - 2e^t + C$$

$$\Rightarrow u e^{x^2/2} = x^2 e^{x^2/2} - 2e^{x^2/2} + C$$

$$\Rightarrow u = x^2 - 2 + Ce^{-x^2/2}$$

$$\Rightarrow \tan y = x^2 - 2 + Ce^{-x^2/2}$$

$$\text{Ex. 2.8.6 : Solve } \frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y.$$

GTU : Summer-20, Maths-2, Marks 3

Sol. :

$$\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$$

$$\sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$$

$$\text{Let } u = \tan y$$

$$\frac{du}{dx} = \sec^2 y \frac{dy}{dx}$$

$$\therefore \frac{du}{dx} + 2xu = x^3$$

$$\text{Here, } P(x) = 2x, Q(x) = x^3$$

$$\text{I.F.} = e^{\int P(x) dx} = e^{\int 2x dx} = e^{x^2}$$

Solution is

$$u \cdot (\text{I.F.}) = \int Q(x)(\text{I.F.}) dx + C$$

$$\Rightarrow u \cdot e^{x^2} = \int x^3 \cdot e^{x^2} dx + C$$

$$\text{Let } x^2 = t$$

$$\Rightarrow 2x dx = dt$$

$$\Rightarrow x dx = \frac{dt}{2}$$

$$\therefore u e^{x^2} = \int x^2 \cdot e^{x^2} x dx + C$$

$$\Rightarrow u e^{x^2} = \int t e^t \frac{dt}{2} + C$$

$$\Rightarrow u e^{x^2} = \frac{t}{2} e^t - \int \frac{1}{2} e^t dt + C$$

$$\Rightarrow u e^{x^2} = \frac{t}{2} e^t - \frac{1}{2} e^t + C$$

$$\Rightarrow u e^{x^2} = \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} + C$$

$$\Rightarrow \tan y = \frac{x^2}{2} - \frac{1}{2} + Ce^{-x^2}$$

$$\text{Ex. 2.8.7 : Solve } \frac{dy}{dx} + \frac{y}{x} \log y = \frac{y}{x^2} (\log y)^2.$$

GTU : Winter-19, Maths-3, Marks 4

Sol. : Divide both sides by $y(\log y)^2$,

$$\Rightarrow \frac{1}{y(\log y)^2} \frac{dy}{dx} + \frac{1}{\log y} \frac{1}{x} = \frac{1}{x^2}$$

$$\text{Let } u = \frac{1}{\log y}$$

$$\Rightarrow \frac{du}{dx} = -\frac{1}{y(\log y)^2} \frac{dy}{dx}$$

$$\Rightarrow -\frac{du}{dx} = \frac{1}{y(\log y)^2} \frac{dy}{dx}$$

$$\therefore -\frac{du}{dx} + \frac{u}{x} = \frac{1}{x^2}$$

$$\Rightarrow \frac{du}{dx} - \frac{1}{x} u = -\frac{1}{x^2}$$

$$\text{Here, } P(x) = -\frac{1}{x}, Q(x) = -\frac{1}{x^2}$$

$$\text{I.F.} = e^{\int P(x) dx} = e^{\int -\frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$$

$\therefore$ Solution is

$$u \cdot (\text{I.F.}) = \int Q(x)(\text{I.F.}) dx + C$$

$$\Rightarrow u \cdot \frac{1}{x} = \int \left(-\frac{1}{x^2}\right) \left(\frac{1}{x}\right) dx + C$$

$$\Rightarrow \frac{u}{x} = -\int \frac{1}{x^3} dx + C$$

$$\Rightarrow \frac{u}{x} = \frac{x^{-2}}{2} + C$$

$$\Rightarrow \frac{1}{x \log y} = \frac{1}{2x^2} + C$$

Ex. 2.8.8 : Solve $\frac{dy}{dx} = \frac{y^3}{e^{2x} + y^2}$.

GTU : Winter-19, Maths-2, Marks 3

Sol. : $\frac{dx}{dy} = \frac{e^{2x} + y^2}{y^3}$

$$\Rightarrow \frac{dx}{dy} = \frac{e^{2x}}{y^3} + \frac{1}{y}$$

$$\Rightarrow \frac{dx}{dy} - \frac{1}{y} = \frac{e^{2x}}{y^3}$$

$$\Rightarrow e^{-2x} \frac{dx}{dy} - e^{-2x} \frac{1}{y} = \frac{1}{y^3}$$

Let $u = e^{-2x}$

$$\frac{du}{dy} = -e^{-2x} \cdot \frac{dx}{dy}$$

$$\Rightarrow -\frac{du}{dy} = e^{-2x} \cdot \frac{dx}{dy}$$

$$\therefore -\frac{du}{dy} - \frac{u}{y} = \frac{1}{y^3}$$

$$\Rightarrow \frac{du}{dy} + \frac{1}{y}u = -\frac{1}{y^3}$$

Here, $P(y) = \frac{1}{y}$, $Q(y) = -\frac{1}{y^3}$

$$\begin{aligned} \text{I.F.} &= e^{\int P(y) dy} = e^{\int \frac{1}{y} dy} \\ &= e^{\log y} = y \end{aligned}$$

$\therefore$ Solution is

$$u \cdot (\text{I.F.}) = \int Q(y)(\text{I.F.}) dy + C$$

$$\Rightarrow u \cdot y = \int \left(-\frac{1}{y^3} \right) y dy + C$$

$$\Rightarrow u y = -\int \frac{1}{y^2} dy + C$$

$$\Rightarrow u y = \frac{1}{y} + C$$

$$\Rightarrow e^{-2x} \cdot y = \frac{1}{y} + C$$

2.9 First Order Differential Equations with Higher Degree

GTU : Winter-21, 22, 23,
Summer-19, 20, 21, 22, 23, 24

In previous sections, we discuss differential equations of first order and first degree. In this section, we shall discuss first order differential equation with higher degree. In this section, we shall use the standard notation for the derivative $\frac{dy}{dx}$ by p .

So, the differential equation of first order and higher degree can be written as $f(x, y, p) = 0$.

There are different types for first order and higher degree differential equations.

- I) Differential equation solvable for p ,
- II) Differential equation solvable for y ,
- III) Differential equation solvable for x ,
- IV) Clairaut's type equation.

Type I : Differential equation solvable for p :

A differential equation of first order and n^{th} degree is of the form.

$$a_n \left(\frac{dy}{dx} \right)^n + a_{n-1} \left(\frac{dy}{dx} \right)^{n-1} + \dots + a_1 \left(\frac{dy}{dx} \right) + a_0 = 0$$

$$\text{or } a_n p^n + a_{n-1} p^{n-1} + \dots + a_1 p + a_0 = 0$$

where, $a_n, a_{n-1}, \dots, a_1, a_0$ are functions of x and y .

As it is a polynomial of degree n in term of p . It can be factorize as

$$(p - \phi_1(x, y)) (p - \phi_2(x, y)) \dots (p - \phi_n(x, y)) = 0$$

Thus, we have n first order first degree differential equation.

$$p = \phi_i(x, y), i = 1, 2, 3, \dots, n$$

If the solution of each equation is $f_i(x, y, C_i) = 0$, then the general solution can be written as

$$f_1(x, y, C_1) \cdot f_2(x, y, C_2) \cdot \dots \cdot f_n(x, y, C_n) = 0$$

As, the equation is of first order, the solution required only one arbitrary constant.

So, consider $C_1 = C_2 = \dots = C_n = C$.

$\therefore$ The general solution can be written as

$$f_1(x, y, C) \cdot f_2(x, y, C) \cdot \dots \cdot f_n(x, y, C) = 0$$

Ex. 2.9.1 : Solve $p^2 + 2p y \cot x = y^2$.

GTU : Summer-22, Maths-2, Marks 4

Sol. : $p^2 + 2y \cot x p - y^2 = 0$

$$\Rightarrow p = \frac{-2y \cot x \pm \sqrt{4y^2 \cot^2 x + 4y^2}}{2}$$

$$\Rightarrow p = \frac{-2y \cot x \pm 2y \operatorname{cosec} x}{2}$$

$$\Rightarrow p = y(-\cot x \pm \operatorname{cosec} x)$$

$$\Rightarrow \frac{dy}{dx} = y(-\cot x + \operatorname{cosec} x)$$

or $\frac{dy}{dx} = y(-\cot x - \operatorname{cosec} x)$

$$\Rightarrow \frac{1}{y} dy = (-\cot x + \operatorname{cosec} x) dx$$

or $\frac{1}{y} dy = (-\cot x - \operatorname{cosec} x) dx$

Integrate both sides, we get

$$\Rightarrow \log y = -\log \sin x + \log (\operatorname{cosec} x - \cot x)$$

or $\log y = -\log \sin x - \log (\operatorname{cosec} x - \cot x)$

$$\Rightarrow \log y = \log \left(\frac{1 - \cos x}{\sin^2 x} + \log C \right)$$

or $\log y = -\log (1 - \cos x) + \log C$

$$\Rightarrow \log y = \log \left(\frac{1}{1 + \cos x} \right) + \log C$$

or $\log y + \log (1 - \cos x) = \log C$

$$\Rightarrow \log y = -\log (1 + \cos x) + \log C$$

or $\log (y(1 - \cos x)) = \log C$

$$\Rightarrow \log (y(1 + \cos x)) = \log C$$

or $\log (y(1 - \cos x)) = \log C$

$$\Rightarrow y(1 + \cos x) - C = 0$$

or $y(1 - \cos x) - C = 0$

$\therefore$ The general solution is

$$(y(1 + \cos x) - C)(y(1 - \cos x) - C) = 0$$

Ex. 2.9.2 : Solve $x^2 p^2 - 4y^2 = 0$.

GTU : Winter-23, Maths-2, Marks 4

Sol. : $x^2 p^2 - 4y^2 = 0$

$$\Rightarrow (xp - 2y)(xp + 2y) = 0$$

$$\Rightarrow xp - 2y = 0 \quad \text{or} \quad xp + 2y = 0$$

$$\Rightarrow p = \frac{2y}{x} \quad \text{or} \quad p = -\frac{2y}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2y}{x} \quad \text{or} \quad \frac{dy}{dx} = -\frac{2y}{x}$$

$$\Rightarrow \frac{1}{y} dy = \frac{2}{x} dx \quad \text{or} \quad \frac{1}{y} dy = -\frac{2}{x} dx$$

Integrate both sides, we get

$$\log y = 2 \log x + \log C$$

or $\log y = -2 \log x + C$

$$\Rightarrow y = x^2 C \quad \text{or} \quad yx^2 = C$$

$$\Rightarrow y - x^2 C = 0 \quad \text{or} \quad yx^2 - C = 0$$

$\therefore$ The general solution is

$$(y - x^2 C)(yx^2 - C) = 0$$

Ex. 2.9.3 : Solve $xyp^2 - (x^2 + y^2)p + xy = 0$.

Sol. : $xyp^2 - (x^2 + y^2)p + xy = 0$

$$\Rightarrow xyp^2 - x^2 p - y^2 p + xy = 0$$

$$\Rightarrow xp(yp - x) - y(yp - x) = 0$$

$$\Rightarrow (yp - x)(xp - y) = 0$$

$$\Rightarrow yp - x = 0 \quad \text{or} \quad xp - y = 0$$

$$\Rightarrow p = \frac{x}{y} \quad \text{or} \quad p = \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x}{y} \quad \text{or} \quad \frac{dy}{dx} = \frac{y}{x}$$

$$\Rightarrow y dy = x dx \quad \text{or} \quad \frac{1}{y} dy = \frac{1}{x} dx$$

$$\Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \quad \text{or} \quad \log y = \log x + \log C$$

$$\Rightarrow y^2 - x^2 - C = 0 \quad \text{or} \quad y - Cx = 0$$

$\therefore$ The general solution is

$$(y^2 - x^2 - C)(y - Cx) = 0$$

Ex. 2.9.4 : Solve $p^2 - yp = x^2 + xy$.

Sol. : $p^2 - x^2 - yp - xy = 0$

$$\Rightarrow (p-x)(p+x) - y(p+x) = 0$$

$$\Rightarrow (p+x)(p-x-y) = 0$$

$$\Rightarrow p+x=0 \quad \text{or} \quad p-x-y=0$$

$$\Rightarrow p = -x \quad \text{or} \quad \frac{dy}{dx} - y = x$$

$$\Rightarrow \frac{dy}{dx} = -x$$

$$\Rightarrow dy = -x \, dx$$

$$\Rightarrow y = -\frac{x^2}{2} + C$$

$$\Rightarrow 2y + x^2 - C = 0$$

$$\text{I.F.} = e^{\int P(x) \, dx} = e^{\int (-1) \, dx} = e^{-x}$$

Solution is

$$ye^{-x} = \int x e^{-x} \, dx + C$$

$$ye^{-x} = -xe^{-x} + \int e^{-x} \, dx + C$$

$$ye^{-x} = -xe^{-x} - e^{-x} + C$$

$$y = -x - 1 + Ce^x$$

$$y + x + 1 - Ce^x = 0$$

$\therefore$ The general solution is

$$(2y + x^2 - C)(y + x + 1 - Ce^x) = 0$$

Type II : Differential equation solvable for y :

If the given first order differential equation is of the form

$$y = f(x, p) \quad \dots (i)$$

then it can be solvable for y.

Differentiating (i) w.r.t. x we get

$$\frac{dy}{dx} = \phi\left(x, p, \frac{dp}{dx}\right)$$

$$\Rightarrow p = \phi\left(x, p, \frac{dp}{dx}\right)$$

This is a differential equation in p and x. Solve this differential equation for p. The general solution is written as

$$\psi(x, p, C) = 0 \quad \dots (ii)$$

where C is a constant

The elimination of p from equations (i) and (ii) gives the general solution.

Ex. 2.9.5 : Solve $y + px = x^4 p^2$.

GTU : Summer-20, 24, Maths-2, Marks 7

Sol. :

$$y + px = x^4 p^2$$

$$\Rightarrow y = -px + x^4 p^2 \quad \dots (i)$$

Differentiating w.r.t. x,

$$\frac{dy}{dx} = -p - x \frac{dp}{dx} + 4x^3 p^2 + x^4 2p \frac{dp}{dx}$$

$$\Rightarrow p = -p - x \frac{dp}{dx} + 4x^3 p^2 + 2px^4 \frac{dp}{dx}$$

$$\Rightarrow 2p + x \frac{dp}{dx} - 4x^3 p^2 - 2px^4 \frac{dp}{dx} = 0$$

$$\Rightarrow 2p \frac{dx}{dp} + x - 4x^3 p^2 \frac{dx}{dp} - 2x^4 p = 0$$

$$\Rightarrow \left(2p \frac{dx}{dp} + x\right) - 2x^3 p \left(2p \frac{dx}{dp} + x\right) = 0$$

$$\Rightarrow \left(2p \frac{dx}{dp} + x\right)(1 - 2x^3 p) = 0$$

As $(1 - 2x^3 p)$ does not contain $\frac{dx}{dp}$, we neglect it

We need to solve

$$2p \frac{dx}{dp} + x = 0$$

$$\Rightarrow \frac{dp}{p} = -2 \frac{dx}{x}$$

Integrating both sides, we get

$$\log p = -2 \log x + \log C$$

$$\Rightarrow px^2 = C$$

$$\Rightarrow p = \frac{C}{x^2} \quad \dots (ii)$$

Eliminating p from (i) and (ii), we get

$$y = \frac{-C}{x^2} x + x^4 \left(\frac{C}{x^2}\right)^2$$

$$\Rightarrow y = \frac{-C}{x} + C^2$$

Ex. 2.9.6 : Solve $y = 2px - xp^2$.

GTU : Winter-22, Maths-2, Marks 3

Sol. :

$$y = 2px - xp^2 \quad \dots (i)$$

Differentiating w.r.t. x ,

$$\frac{dy}{dx} = 2p + 2x \frac{dp}{dx} - p^2 - x^2 p \frac{dp}{dx}$$

$$\Rightarrow p = 2p + 2x \frac{dp}{dx} - p^2 - 2xp \frac{dp}{dx}$$

$$\Rightarrow p + 2x \frac{dp}{dx} - p^2 - 2xp \frac{dp}{dx} = 0$$

$$\Rightarrow p - p^2 + 2x \frac{dp}{dx} - 2xp \frac{dp}{dx} = 0$$

$$\Rightarrow p(1-p) + 2x \frac{dp}{dx}(1-p) = 0$$

$$\Rightarrow (1-p) \left(p + 2x \frac{dp}{dx} \right) = 0$$

As $(1-p)$ does not contain $\frac{dp}{dx}$, we neglect it. $\therefore$ We have to find solution of

$$p + 2x \frac{dp}{dx} = 0$$

$$\Rightarrow 2x \frac{dp}{dx} = -p$$

$$\Rightarrow \frac{1}{p} dp = -\frac{1}{2x} dx$$

Integrating both sides, we get

$$\log p = -\frac{1}{2} \log x + \log C$$

$$\Rightarrow \log p = \log \frac{C}{\sqrt{x}}$$

$$\Rightarrow p = \frac{C}{\sqrt{x}} \quad \dots (ii)$$

Eliminating p from (i) and (ii), we get

$$y = 2x \frac{C}{\sqrt{x}} - x \frac{C^2}{x}$$

$$y = 2C\sqrt{x} - C^2$$

Ex. 2.9.7 : Solve $p^2 - xp + y = 0$.

GTU : Summer-24, Maths-2, Marks 3

Sol. : $p^2 - xp + y = 0$

$$\Rightarrow y = px - p^2$$

Differentiating w.r.t. x

$$\frac{dy}{dx} = p + x \frac{dp}{dx} - 2p \frac{dp}{dx}$$

$$\Rightarrow p = p + (x - 2p) \frac{dp}{dx}$$

$$\Rightarrow (x - 2p) \frac{dp}{dx} = 0$$

As $(x - 2p)$ does not contain $\frac{dp}{dx}$, we neglect it.

$$\therefore \frac{dp}{dx} = 0$$

$$\Rightarrow p = C$$

Eliminating p from (i) and (ii), we get

$$y = Cx - C^2$$

Ex. 2.9.8 : Solve $xp^2 - 2py + 4x = 0$.

Sol. :

$$xp^2 - 2py + 4x = 0$$

$$\Rightarrow 2py = xp^2 + 4x$$

$$\Rightarrow y = \frac{xp^2 + 4x}{2p}$$

$$\Rightarrow y = x \left(\frac{p}{2} + \frac{2}{p} \right)$$

Differentiating w.r.t. x ,

$$\frac{dy}{dx} = \left(\frac{p}{2} + \frac{2}{p} \right) + x \left(\frac{1}{2} - \frac{2}{p^2} \right) \frac{dp}{dx}$$

$$\Rightarrow p - \left(\frac{p}{2} + \frac{2}{p} \right) = x \left(\frac{1}{2} - \frac{2}{p^2} \right) \frac{dp}{dx}$$

$$\Rightarrow \left(\frac{p}{2} - \frac{2}{p} \right) = x \left(\frac{1}{2} - \frac{2}{p^2} \right) \frac{dp}{dx}$$

$$\Rightarrow \left(\frac{p^2 - 4}{2p} \right) = x \left(\frac{p^2 - 4}{2p^2} \right) \frac{dp}{dx}$$

$$\Rightarrow \frac{1}{x} dx = \left(\frac{p^2 - 4}{2p^2} \right) \left(\frac{2p}{p^2 - 4} \right) dp$$

$$\Rightarrow \frac{1}{x} dx = \frac{1}{p} dp$$

Integrating both sides, we get

$$\Rightarrow \log p = \log x + \log C$$

$$\Rightarrow p = Cx \quad \dots (ii)$$

Eliminating p from (i) and (ii), we get

$$x(C^2 x^2) - 2y(Cx) + 4x = 0$$

$$\Rightarrow C^2 x^3 - 2xyC + 4x = 0$$

$$\Rightarrow C^2 x^2 - 2yC + 4 = 0$$

Type III : Differential equation solvable for x :

If the given first order differential equation is of the form.

$$x = f(y, p) \quad \dots (i)$$

then it can be solvable for x.

Differentiating (i) w.r.t. y, we get

$$\frac{dx}{dy} = \phi \left(y, p, \frac{dp}{dy} \right)$$

$$\Rightarrow \frac{1}{p} = \phi \left(y, p, \frac{dp}{dy} \right)$$

This is a differential equation in p and y. Solve this differential equation for p. The general solution is written as

$$\psi(y, p, C) = 0 \quad \dots (ii)$$

where C is constant.

The elimination of p from equation (i) and (ii) gives the general solution.

Ex. 2.9.9 : Solve $y = 2px + y^2 p^3$.

Sol. :

$$y = 2px + y^2 p^3$$

$$\Rightarrow 2px = y - y^2 p^3$$

$$\Rightarrow x = \frac{y}{2p} - \frac{y^2 p^2}{2} \quad \dots (i)$$

Differentiating w.r.t. y,

$$\frac{dx}{dy} = \frac{1}{2p} - \frac{y}{2p^2} \frac{dp}{dy} - yp^2 - y^2 p \frac{dp}{dy}$$

$$\Rightarrow \frac{1}{p} = \frac{1}{2p} - \left(\frac{y}{2p^2} + y^2 p \right) \frac{dp}{dy} - yp^2$$

$$\Rightarrow \frac{1}{2p} + yp^2 + \left(\frac{y}{2p^2} + y^2 p \right) \frac{dp}{dy} = 0$$

$$\Rightarrow \left(\frac{1}{2p} + yp^2 \right) + \left(\frac{1}{2p} + yp^2 \right) \frac{y}{p} \frac{dp}{dy} = 0$$

$$\Rightarrow \left(\frac{1}{2p} + yp^2 \right) + \left(1 + \frac{y}{p} \frac{dp}{dy} \right) = 0$$

As $\left(\frac{1}{2p} + yp^2 \right)$ does not contain $\frac{dp}{dy}$, we neglect it.

$$\therefore 1 + \frac{y}{p} \frac{dp}{dy} = 0$$

$$\Rightarrow \frac{1}{p} dp = -\frac{1}{y} dy$$

$$\Rightarrow \log p = -\log y + \log C$$

$$\Rightarrow p = C/y \quad \dots (ii)$$

Eliminating p from (i) and (ii), we get

$$x = \frac{y^2}{2C} - \frac{y^2}{2} \frac{C^2}{y^2}$$

$$\Rightarrow x = \frac{y^2}{2C} - \frac{C^2}{2}$$

Ex. 2.9.10 : Solve $xp^2 - 2py + x = 0$.

Sol. : $xp^2 - 2py + x = 0$

$$\Rightarrow x(p^2 + 1) = 2py$$

$$\Rightarrow x = \frac{2py}{p^2 + 1} \quad \dots (i)$$

Differentiating w.r.t. y,

$$\frac{dx}{dy} = \frac{2p}{p^2 + 1} + y \left(\frac{2(p^2 + 1) - 2p(2p)}{(p^2 + 1)^2} \right) \frac{dp}{dy}$$

$$\Rightarrow \frac{1}{p} = \frac{2p}{p^2 + 1} + y \left(\frac{2p^2 + 2 - 4p^2}{(p^2 + 1)^2} \right) \frac{dp}{dy}$$

$$\Rightarrow \frac{1}{p} - \frac{2p}{p^2+1} = y \left(\frac{2(1-p^2)}{(p^2+1)^2} \right) \frac{dp}{dy}$$

$$\Rightarrow \frac{p^2+1-2p^2}{p(p^2+1)} = y \left(\frac{2(1-p^2)}{(p^2+1)^2} \right) \frac{dp}{dy}$$

$$\Rightarrow \frac{1-p^2}{p(p^2+1)} = y \left(\frac{2(1-p^2)}{(p^2+1)^2} \right) \frac{dp}{dy}$$

$$\Rightarrow \frac{1}{y} dy = \frac{2p}{p^2+1} dp$$

Integrating both sides, we get

$$\log y = \log(p^2+1) + \log C$$

$$\Rightarrow y = (p^2+1) \cdot C$$

$$\text{or } p^2+1 = Cy$$

$$\Rightarrow p^2 = Cy - 1$$

$$\Rightarrow p = \sqrt{Cy-1} \quad \dots (ii)$$

Eliminating p from (i), we get

$$x = \frac{2y\sqrt{Cy-1}}{Cy}$$

$$\Rightarrow Cxy = 2y\sqrt{Cy-1}$$

$$\text{Ex. 2.9.11 : Solve } p = \tan \left(x - \frac{p}{1+p^2} \right).$$

Sol. :

$$p = \tan \left(x - \frac{p}{1+p^2} \right)$$

$$\Rightarrow \tan^{-1} p = x - \frac{p}{1+p^2}$$

$$\Rightarrow x = \frac{p}{1+p^2} + \tan^{-1} p \quad \dots (i)$$

Differentiating w.r.t. y,

$$\frac{dx}{dy} = \frac{(1+p^2) - p(2p)}{(1+p^2)^2} \left(\frac{dp}{dy} \right) + \frac{1}{1+p^2} \frac{dp}{dy}$$

$$\Rightarrow \frac{1}{p} = \frac{1-p^2}{(1+p^2)^2} \frac{dp}{dy} + \frac{1}{(1+p^2)} \frac{dp}{dy}$$

$$\Rightarrow \frac{1}{p} = \frac{2}{(1+p^2)^2} \frac{dp}{dy}$$

$$\Rightarrow dy = \frac{2p}{(1+p^2)^2} dp$$

Integrating on both sides,

$$y = -\frac{1}{1+p^2} + C$$

From equation (i) and (ii),

$$x = \frac{p}{1+p^2} + \tan^{-1} p$$

and

$$y = -\frac{1}{1+p^2} + C$$

which are parametric solution of the given differential equation.

Ex. 2.9.12 : Solve $p^2 x^2 = x^2 + p^2$.

GTU : Summer-23, Maths-2, Marks 4

$$\text{Sol. : } p^2 x^2 = x^2 + p^2$$

$$\Rightarrow p^2(x^2 - 1) = x^2$$

$$\Rightarrow p^2 = \frac{x^2}{x^2 - 1}$$

$$\Rightarrow p = \pm \frac{x}{\sqrt{x^2 - 1}}$$

$$p = \frac{x}{\sqrt{x^2 - 1}} \quad \text{or} \quad p = -\frac{x}{\sqrt{x^2 - 1}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x}{\sqrt{x^2 - 1}} \quad \text{or} \quad \frac{dy}{dx} = -\frac{x}{\sqrt{x^2 - 1}}$$

$$\Rightarrow dy = \frac{x}{\sqrt{x^2 - 1}} dx$$

$$\text{or } dy = -\frac{x}{\sqrt{x^2 - 1}} dx$$

$$\Rightarrow dy = \frac{1}{2} \frac{2x}{\sqrt{x^2 - 1}} dx$$

$$\text{or } dy = -\frac{1}{2} \frac{2x}{\sqrt{x^2 - 1}} dx$$

$$\Rightarrow y = \frac{1}{2} \frac{(x^2 - 1)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$\text{or } y = -\frac{1}{2} \frac{(x^2 - 1)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$\Rightarrow y = \sqrt{x^2 - 1} + C \quad \text{or} \quad y = -\sqrt{x^2 - 1} + C$$

$$\Rightarrow y - \sqrt{x^2 - 1} - C = 0 \quad \text{or} \quad y + \sqrt{x^2 - 1} - C = 0$$

$\therefore$ The general solution is

$$(y - \sqrt{x^2 - 1} - C)(y + \sqrt{x^2 - 1} - C) = 0$$

Type IV : Clairaut's Equation :

The first order equation of the form $y = px + f(p)$ is known as Clairaut's equation.

Clairaut's equations is a first order differential equation solvable for y .

Solving equation, we get

$$p = C$$

$\therefore y = Cx + f(C)$ is solution of Clairaut's equation.

Ex. 2.9.13 : Solve $\sin(y - xp) = p$.

**GTU : Summer-19, Maths-3, Marks 4,
Summer-21, Winter-23, Maths-2**

Sol. : $\sin(y - xp) = p$

$$\Rightarrow y - xp = \sin^{-1} p$$

$$\Rightarrow y = px + \sin^{-1} p$$

which is Clairaut's equation.

$\therefore$ The general solution is

$$y = Cx + \sin^{-1} C$$

Ex. 2.9.14 : Solve $(y - px)(p^2 + 1) = \tan^{-1} p$.

GTU : Summer-23, Maths-2, Marks 3

Sol. : $(y - px)(p^2 + 1) = \tan^{-1} p$

$$\Rightarrow y - px = \frac{\tan^{-1} p}{p^2 + 1}$$

$$\Rightarrow y = px + \frac{\tan^{-1} p}{p^2 + 1}$$

which is Clairaut's equation.

$\therefore$ The general solution is

$$y = Cx + \frac{\tan^{-1} C}{C^2 + 1}$$

Ex. 2.9.15 : Solve $(y - px)(p - 1) = p$.

GTU : Winter-21, Maths-2, Marks 3

Sol. : $(y - px)(p - 1) = p$

$$\Rightarrow y - px = \frac{p}{p - 1}$$

$$\Rightarrow y = px + \frac{p}{p - 1}$$

which is Clairaut's equation.

$\therefore$ The general solution is

$$y = Cx + \frac{C}{C - 1}$$

Ex. 2.9.16 : Solve $p = \log(px - y)$.

GTU : Winter-22, Maths-2, Marks 4

Sol. :

$$p = \log(px - y)$$

$$\Rightarrow e^p = px - y$$

$$\Rightarrow y = px - e^p$$

which is Clairaut's equation

$\therefore$ The general solution is

$$y = Cx - e^C$$

Ex. 2.9.17 : Solve $4xp^2 = (3x - a)^2$.

GTU : Winter-21, Maths-2, Marks 4

Sol. :

$$4xp^2 = (3x - a)^2$$

$$\Rightarrow p^2 = \frac{(3x - a)^2}{4x}$$

$$\Rightarrow p = \pm \frac{3x - a}{2\sqrt{x}}$$

$$\Rightarrow p = \pm \left(\frac{3}{2} \sqrt{x} - \frac{a}{2\sqrt{x}} \right)$$

$$p = \frac{3}{2}\sqrt{x} - \frac{a}{2\sqrt{x}}$$

or

$$p = -\frac{3}{2}\sqrt{x} - \frac{a}{2\sqrt{x}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{3}{2}\sqrt{x} - \frac{a}{2\sqrt{x}}$$

or

$$\frac{dy}{dx} = -\frac{3}{2}\sqrt{x} + \frac{a}{2\sqrt{x}}$$

$$\Rightarrow dy = \left(\frac{3}{2}\sqrt{x} - \frac{a}{2\sqrt{x}} \right) dx$$

or

$$dy = \left(-\frac{3}{2}\sqrt{x} + \frac{a}{2\sqrt{x}} \right) dx$$

$$\Rightarrow y = x^{3/2} - a\sqrt{x} + C$$

or

$$y = -x^{3/2} + a\sqrt{x} + C$$

$$\Rightarrow y - x^{3/2} + a\sqrt{x} - C = 0 \quad \text{or} \quad y + x^{3/2} - a\sqrt{x} - C = 0$$

∴ The general solution is

$$(y - x^{3/2} + a\sqrt{x} - C)(y + x^{3/2} - a\sqrt{x} - C) = 0$$